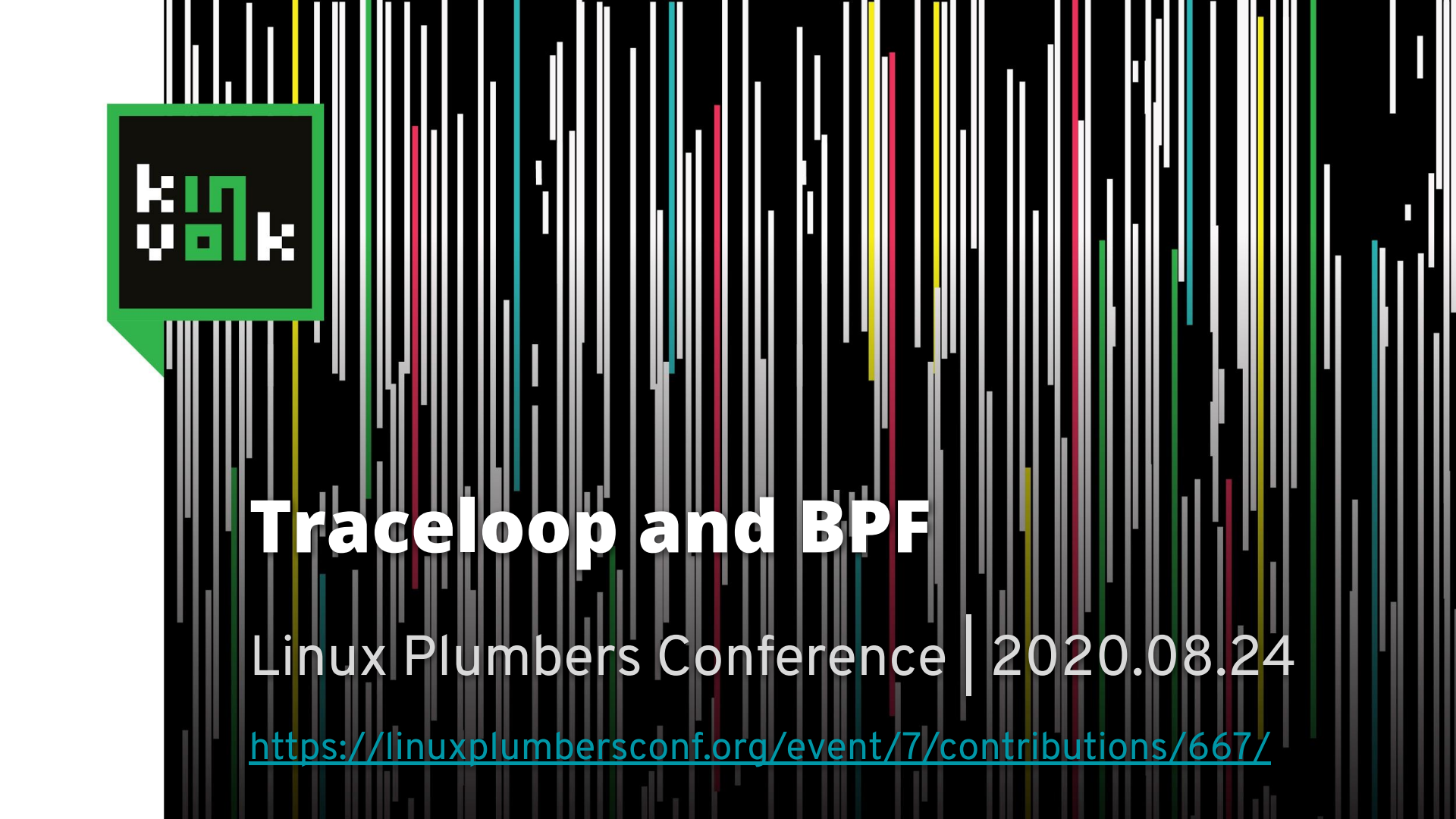

Traceloop and BPF

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https://linuxplumbersconf.org/event/7/contributions/667/

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Traceloop

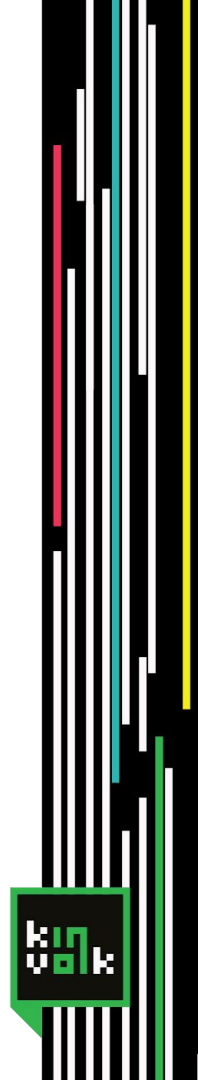
Traces system calls per cgroup, using BPF and overwritable ring buffers to keep the logs

https://github.com/kinvolk/traceloop

Used in Inspektor Gadget

A Collection of BPF gadgets for developers of Kubernetes applications

https://github.com/kinvolk/inspektor-gadget



Problem statement

- Debugging distributed applications is hard
- Tracing tools can help us to see what's going on
- `strace` is great but needs to attach to every processes before the event happens

  - ❏ Would need to know when crashes happen
  - ❏ Performance impact to high if always-on

Idea: “Flight Recorder”

- Capture syscalls with BPF instead of `ptrace`
- Save the events to a per-cgroup (or similar) ring buffer without leaving kernel space
- Assume that discarding old events is acceptable
- Only transfer the ring buffer events to userspace when requested
- Limit global impact by only tracing a list of cgroups (no-op otherwise)

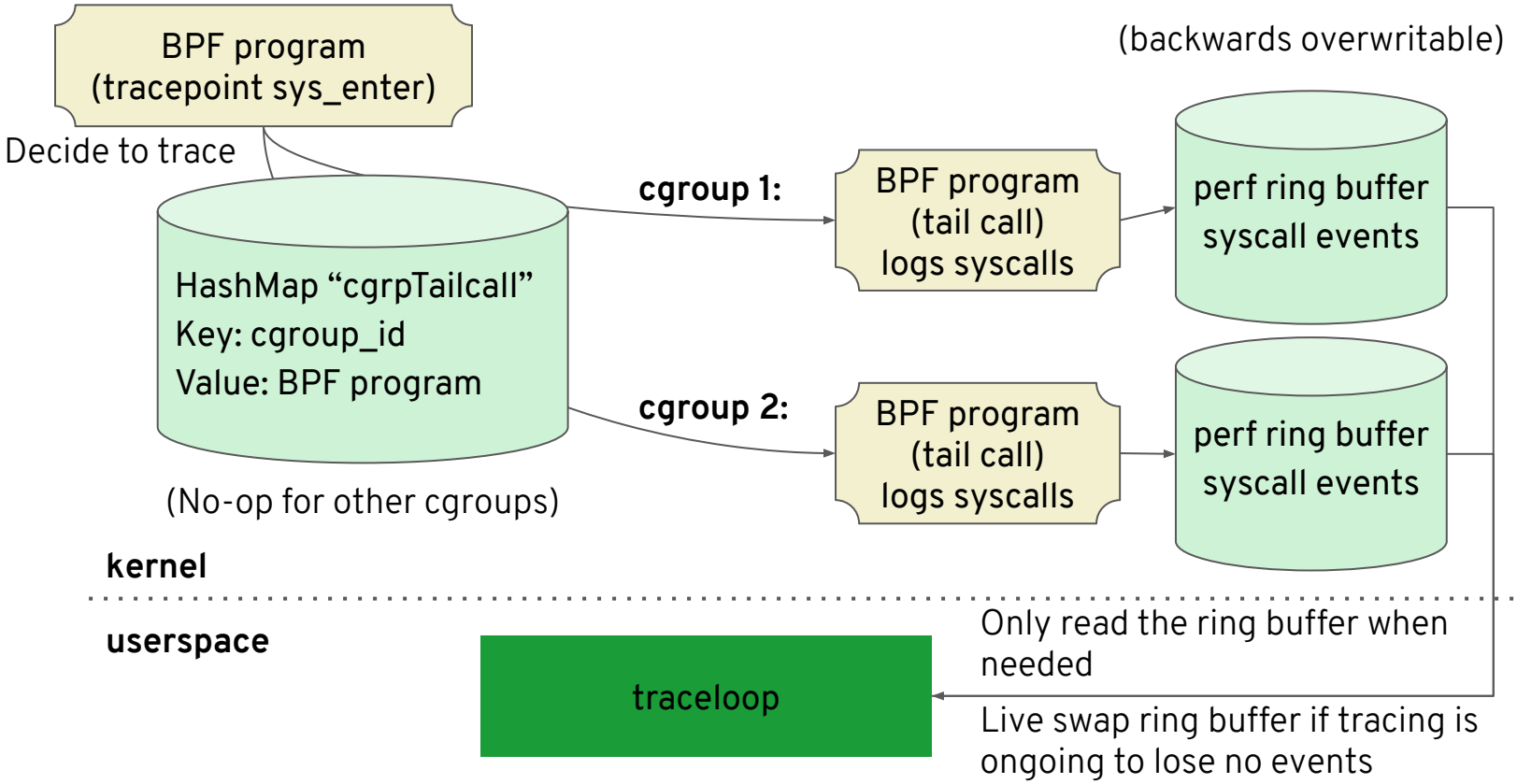
Comparing strace and traceloop

	strace	traceloop
Capture method	ptrace	BPF on tracepoints
Granularity	process	global, filter cgroup/UTS namespace[1]
Speed	slow	fast
Reliability	Synchronous Cannot lose events	Asynchronous Can lose events Can fail to read buffers (EFAULT) when dumping syscall arguments
Maturity	Covers many cases	Not all syscall arguments are read Only AMD64 syscalls recognized

[1] could also be process

Architecture

Architecture



Different modes of execution (1/5)

1. Tracing a cgroup on the command line
    a. Output dump at exit

```
$ sudo traceloop cgroups --dump-on-exit
                              /sys/fs/cgroup/unified/system.slice/sshd.service
^C
(output)
```

    b. Continuous output (might lose events)

```
$ sudo traceloop cgroups /sys/fs/cgroup/unified/system.slice/sshd.service
(continuous output)
Press Ctrl-S to pause, Ctrl-Q to continue, Ctrl-C to quit
```

Different modes of execution (2/5)

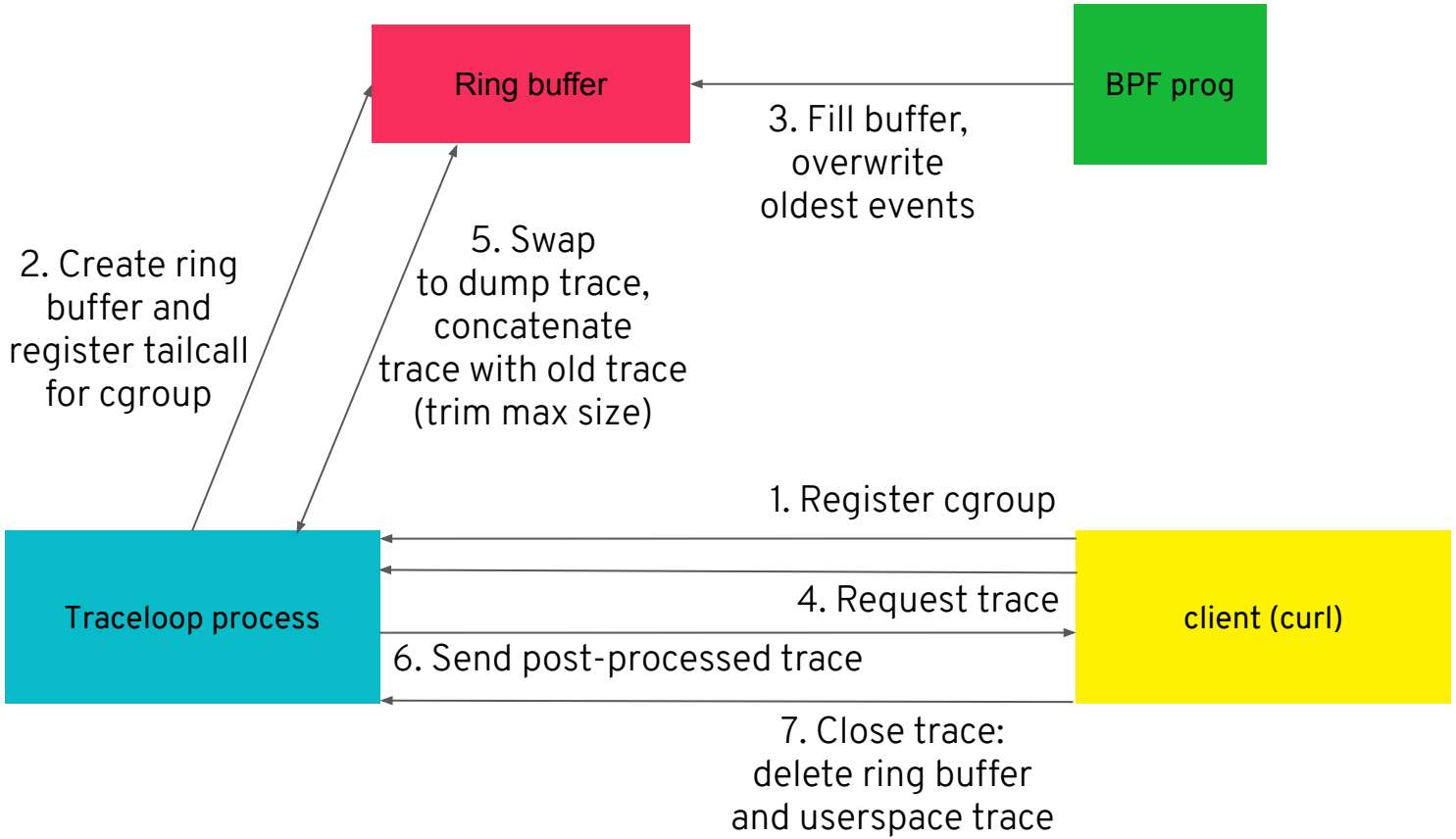
2. Daemon receiving http commands on a unix socket

```
$ sudo traceloop serve &

$ sudo curl --unix-socket /run/traceloop.socket \
     'http://localhost/add?
     name=sshd&cgrouppath=/sys/fs/cgroup/unified/system.slice/sshd.service'
```

- Possible commands in the URL:
  - /add
  - /list
  - /dump, /dump-pod, /dump-by-name, /dump-by-cgroup, /dump-by-traceid
  - /close, /close-by-name

Registration and trace lifetime



Different modes of execution (3/5)

3. Registering a Systemd service:

```
[Service]
ExecStartPre=/bin/sh -c 'curl --unix-socket /run/traceloop.socket
"http://localhost/add?name=myservice&cgrouppath=$(./current-cgroup)"'
ExecStart=/usr/bin/myservice
```

(`contrib/current-cgroup` is available in the traceloop repo)

Different modes of execution (4/5)

4. Kubernetes mode

Inspektor Gadget has a DaemonSet with an entrypoint.sh containing:

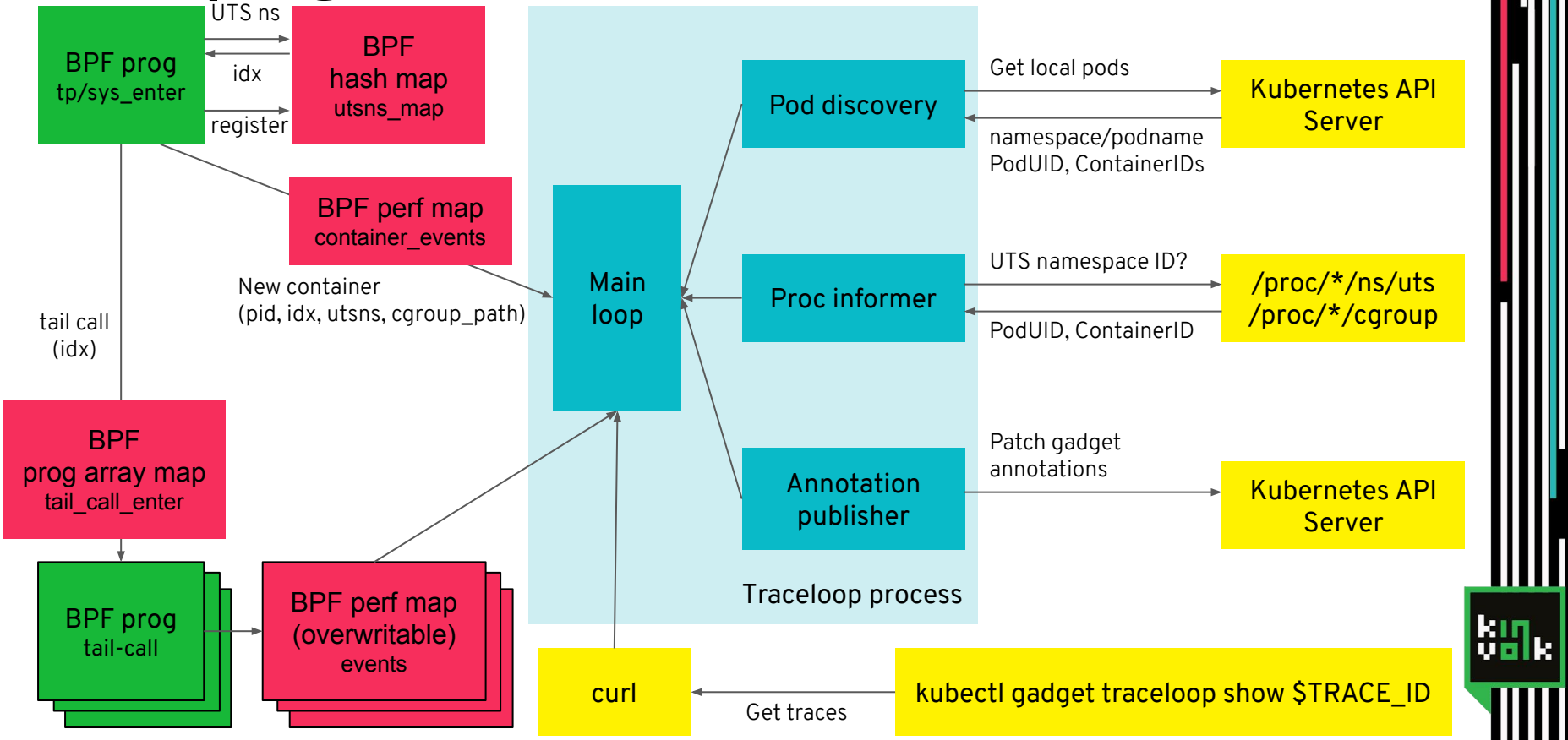
```
/bin/traceloop k8s
```

Include HTTP server on the unix socket as well, plus:

- Pod discovery: fetch local pods from the Kubernetes API Server
- Proc informer: regularly look in /proc/
- Annotation publisher

On Pod termination the BPF ring buffer is dumped and recycled, the trace is kept in userspace for 3 hours

Keeping track of Kubernetes containers



Different modes of execution (5/5)

5. Start traceloop as container

Same possibilities as using the traceloop binary but in a container

```
$ docker run --rm \
    -v /sys/kernel/debug:/sys/kernel/debug \
    -v /sys/fs/cgroup:/sys/fs/cgroup \
    -v /sys/fs/bpf:/sys/fs/bpf \
    -v /run:/run \
    --privileged \
    kinvolk/traceloop
```

How to get a list of syscalls?

Syscalls and their numbers: `/usr/include/asm/unistd_64.h`

Parameters:
`/sys/kernel/debug/tracing/events/syscalls/sys_enter_${name}/format`

Post process some syscalls:

- new{uname,fstat,lstat,stat}→{uname,fstat,lstat,stat}, sendfile64→sendfile, sysctl→_sysctl, umount→umount2

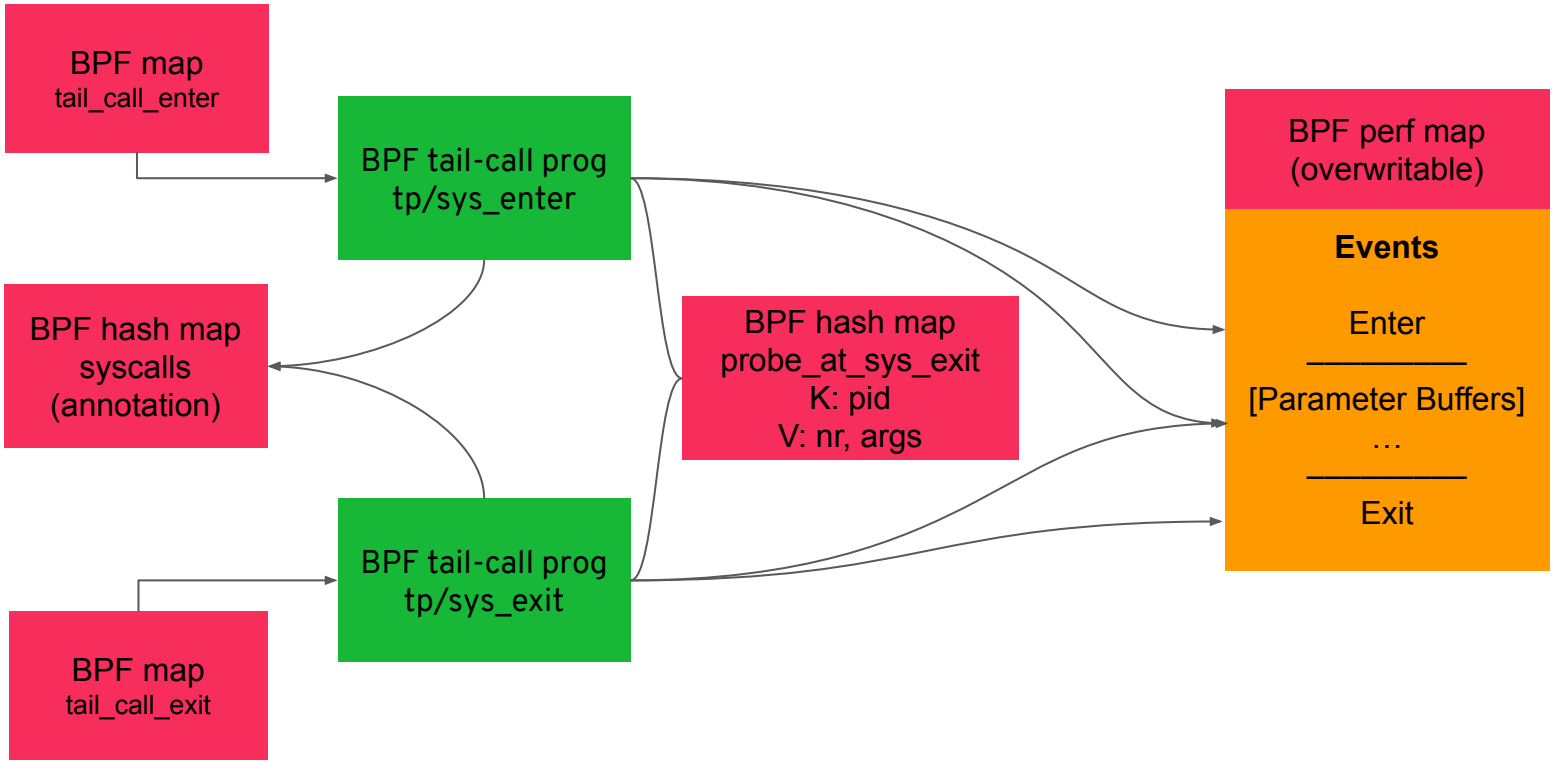
Processing syscall arguments

```
openat(AT_FDCWD, "/etc/ld.so.cache", O_RDONLY|O_CLOEXEC)
write(1, "foo", 3) = 3
read(0, "foo", 131072) = 3
exit_group()
```

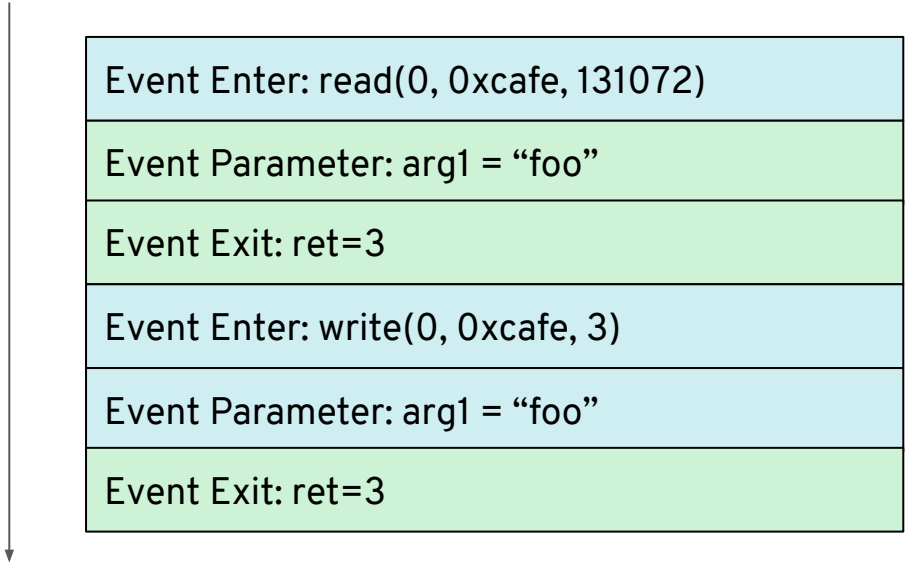
- Dereferencing strings with the correct size
  - NULL terminated strings, size known at sys_enter or known at sys_exit
  - Dereference at sys_enter or sys_exit
- Some syscalls don't return: exit_group()
- Some syscalls return twice: clone()

Syscall	arg0	arg1	arg2	...
opentat()	0	Read until NULL byte	0	0
write()	0	Param length in arg2	0	0
read()	0	Param length in ret, probe at exit	0	0

Processing syscall data



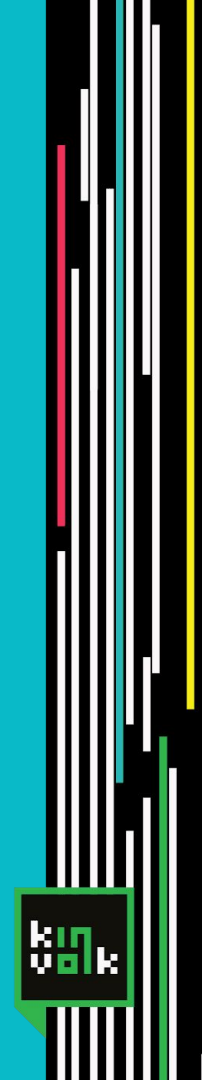
Examples of event stream



Event from sys_enter

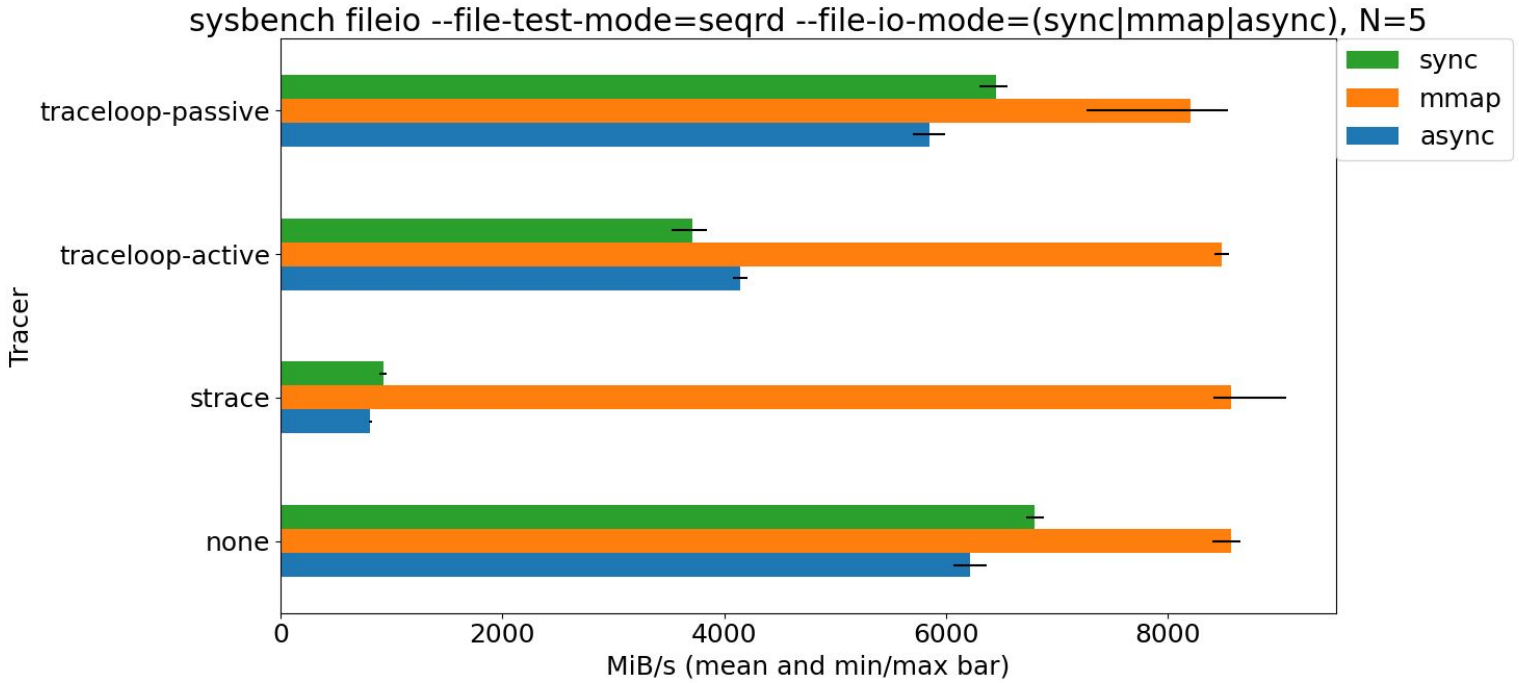
Event from sys_exit

Benchmarks



Benchmark

sync: log buffers, async: log pointers, mmap: no syscalls



Limitations

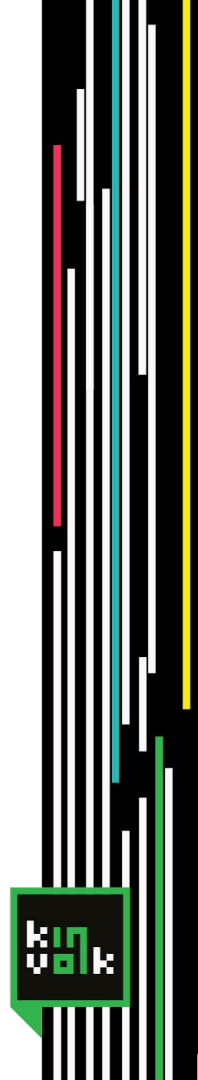
Limitations: bpf_probe_read

```
int main() {
#define FILENAME "/tmp/traceloop-test.data"
  open(FILENAME, O_RDWR, 0);
  open(FILENAME, O_RDWR, 0);
  return 0;
}
```

```
openat(4294967196, "(Pointer deref failed!)", 2, 0) = 3
openat(4294967196, "/tmp/traceloop-test.data", 2, 0) = 4
```

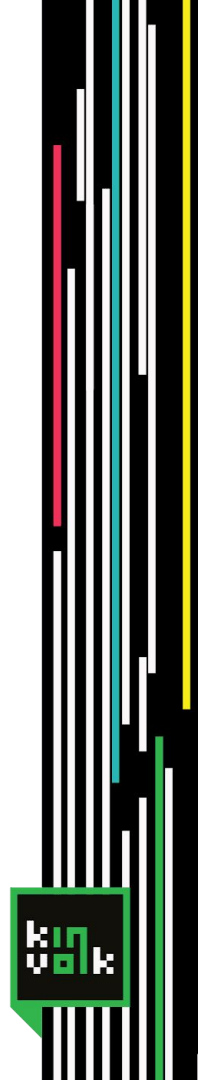
Limitations: BPF

- Memory pages may not be present when dereferencing a syscall parameter: data may not be loaded to memory at the beginning of the syscall
- Cases: First reference to data segment in binary or a reference into an mmaped file
- Workarounds can be to try again at syscall exit but that is not valid for all parameters and in general swapping/cache flushing can also happen



Limitations: Traceloop

- Doesn’t yet detect buffer wraps to mark missing events (even harder when merging per-CPU buffers)
- Merging per-CPU buffers does sorting of timestamps which can have corner cases
- The whole trace data is post-processed each time and not only the new events with those old events they refer to
- Currently max. buffer length for syscall parameters is 128 bytes
- Not all syscalls have annotation for dereferencing parameters and detecting constants or structs



Limitations: Detection of new cgroups/containers

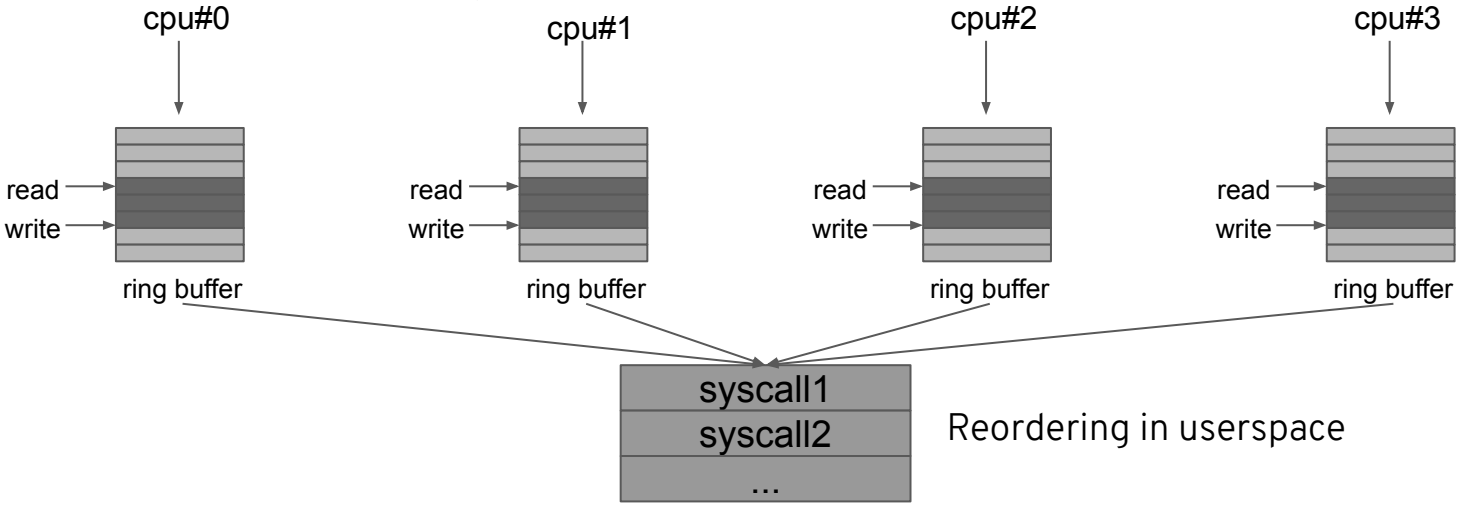
- Possible with systemd `StartPre` trick if permissions are right
- On Kubernetes automatic detection currently done with runc mount tracepoint hack and a pool of free buffers
  - Currently traces all pods, no configuration yet
  - Integration with container OCI hooks was tried with Flatcar Edge but relied on runc patches

Limitations: Memory usage

- Currently per-CPU perf buffers for a cgroup
- Even if buffers are closed, dumps in userspace consume memory too until they are closed (no compression yet)
- In the Kubernetes setup, traceloop's settings:
  - Ring buffer pre-allocation for 128 containers
  - Traceloop uses 64 pages per ring buffer
- Scenario with 16 CPUs:
  - 64 * 4KiB * 16 * 128 = 512 MiB of non-swappable memory

Perf ring buffers: per CPU

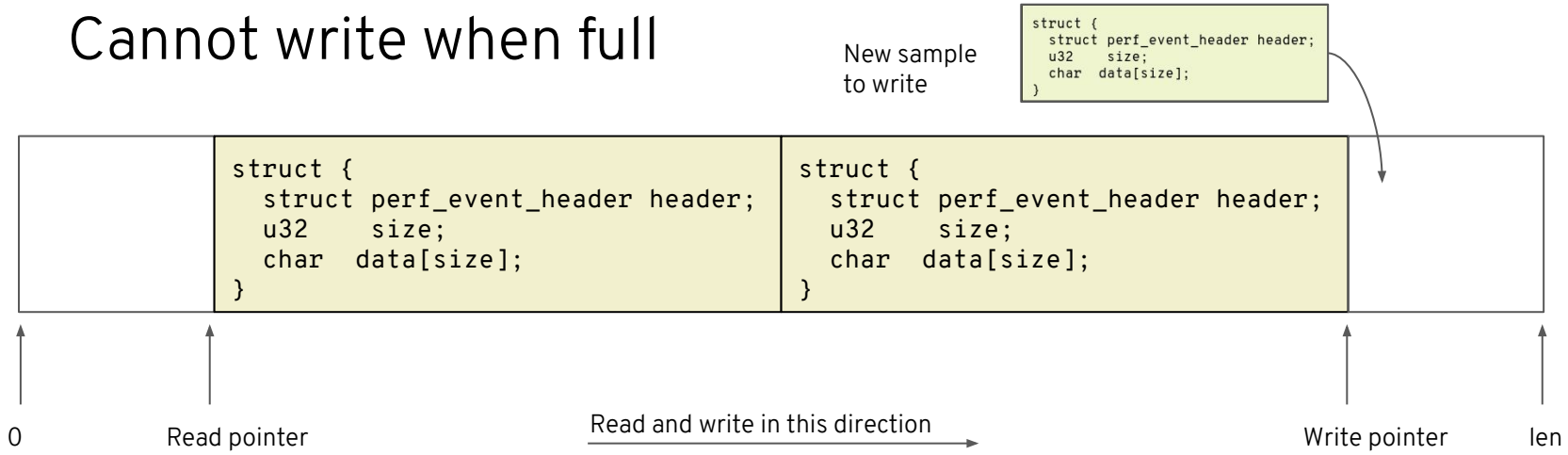
- Map type: BPF_MAP_TYPE_PERF_EVENT_ARRAY
- Linux 4.3
- Lockless writing



Normal perf ring buffers

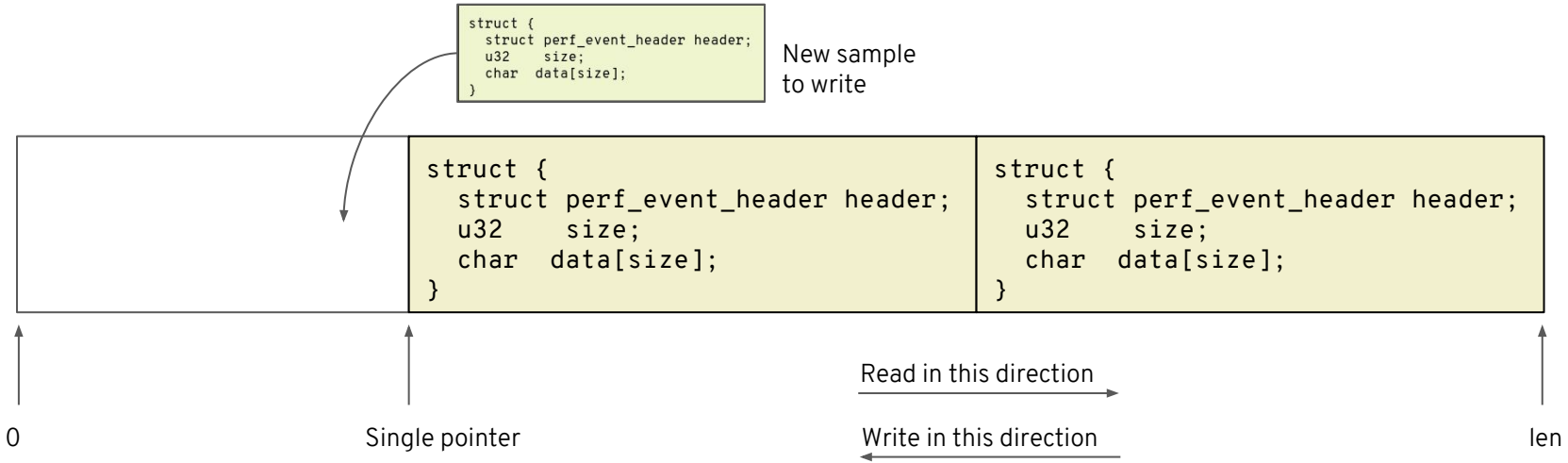
```
struct perf_event_attr attr = {0,};
fd = perf_event_open_map(&attr, ...);
base = mmap(fd, 0, size, PROT_READ|PROT_WRITE, MAP_SHARED);
```

Cannot write when full



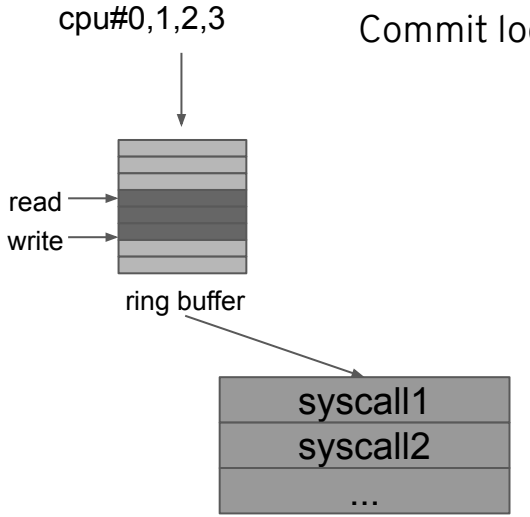
Backward & overwritable ring buffers

```
struct perf_event_attr attr = {0,};
attr.write_backward = 1; /* backward */
fd = perf_event_open_map(&attr, ...);
base = mmap(fd, 0, size, PROT_READ /* overwriteable */, MAP_SHARED);
```



New BPF ring buffers

- Map type: BPF_MAP_TYPE_RINGBUF
- Linux 5.8



Reservation under spinlock
Commit lockless

No reordering in userspace

Decoding arguments with BTF

```
openat(4294967196, "/tmp/traceloop-test.data", 2, 0) = 4
recvfrom(3, 94708461554656, 102400, 0, 0, 0) = 362
```

- Not decoded today:
  - Constants (e.g. AT_FDCWD)
  - Structs (struct msghdr)
- BTF

Outlook: What can be improved?

- Use the new BPF ring buffers to simplify the post-processing and reduce memory usage
  - Need a mode for being backwards overwritable
- Integrate better with systemd
  - A `PrivilegedStartPre`/`StopPostHook` (?) that knows the cgroup path would help to register the service to traceloop
- Anticipate availability of OCI hooks
- Annotate more syscalls/use BTF?

Questions?

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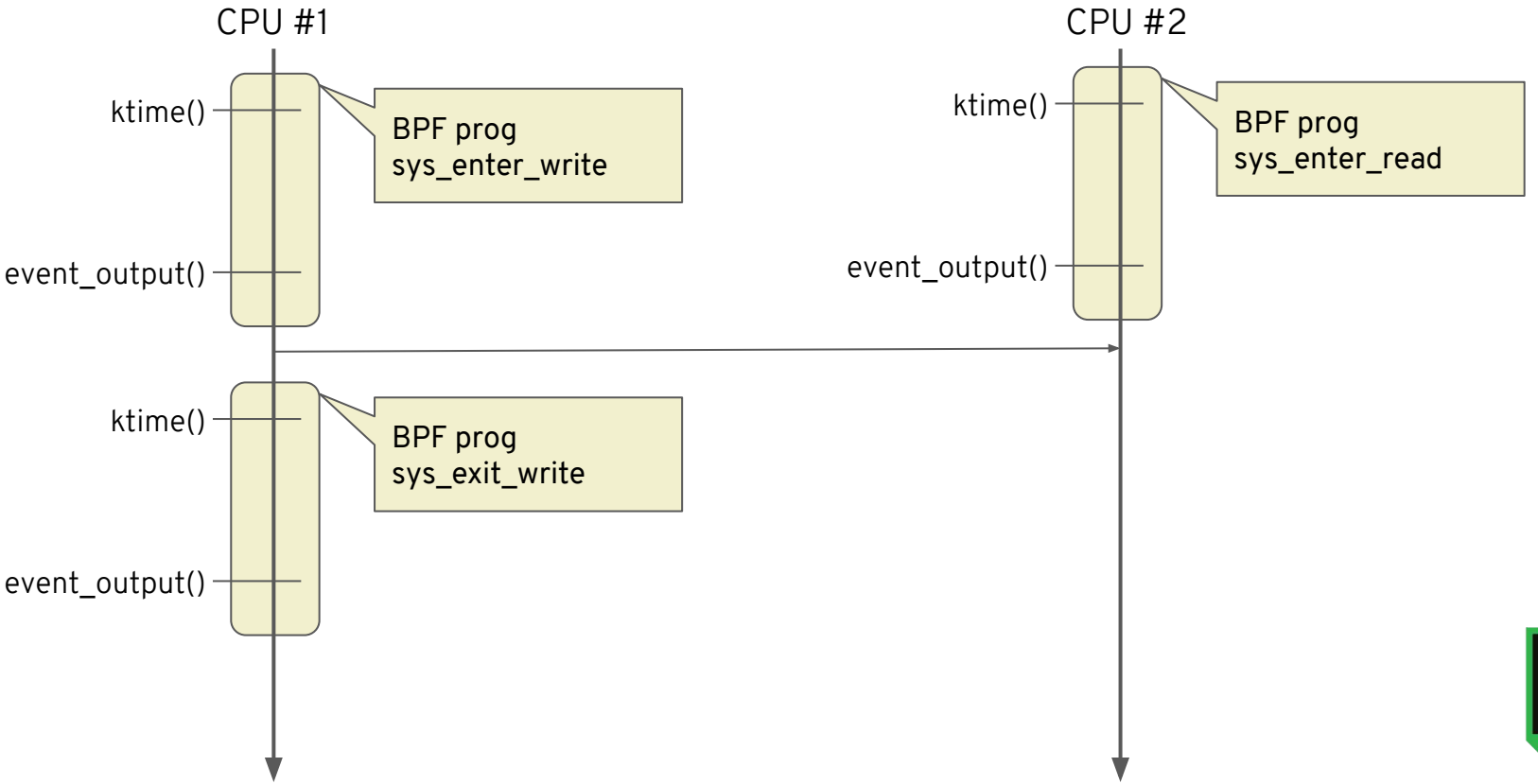
Email: **hello@kinvolk.io**

Kubernetes Slack: #inspektor-gadget
Source Code: https://github.com/kinvolk/traceloop/

Backup slides


CPU #1
CPU #2
ktime()
BPF prog
sys_enter_write
event_output()
ktime()
BPF prog
sys_enter_read
event_output()
ktime()
BPF prog
sys_exit_write
event_output()